

What is Transfer Learning?

Transfer learning is a popular training technique used in deep learning where models that have been trained for a particular task are reused as a base or starting point for another model that is deemed to perform some other task.

Why is it required?

To train an Image classifier that will achieve near or above human level accuracy on Image classification, we will need massive amounts of data, large computational power, and lots of time on our hands. Since this is a big problem for people with little or no resources, some smart researchers

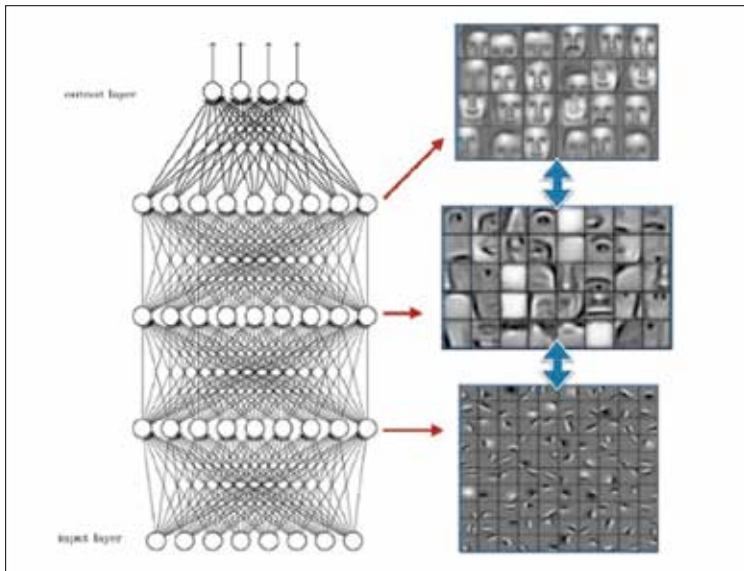


Figure : Neural network learning at increasing levels of complexity

built models, trained on large image datasets like ImageNet, Open Images, and COCO and decided to share their models to the public for reuse. This prevents the need to train an Image classifier from scratch again unless you have a very, very large dataset that is quite different from the ones above and you have large computational power at hand.

Why makes transfer learning work?

Neural Networks learn in an increasingly complex way. This means to say